The Case of Apple's Restrictions on Super Apps: Is the U.S. Department of Justice Right?

Ziheng Yu

The University of Texas at Austin

October 28, 2026

Overview

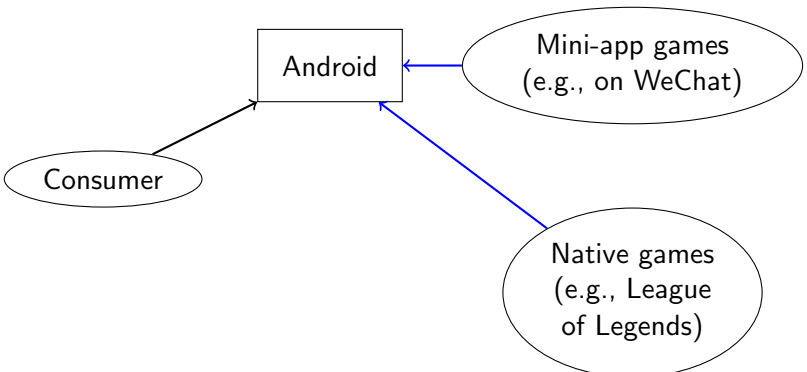
- The U.S. Department of Justice (DOJ) (2024): "Apple has disrupted the growth of apps with broad functionality that would make it easier for consumers to switch between competing smartphone platforms"—which are primarily iOS and Android—by "suppressing super apps." It has
  - "banned apps from categorizing mini programs" and
  - "selectively enforced its contractual rules with developers to prevent developers from monetizing mini programs."
- Apple can effectively restrict iOS apps because it has full control over the iOS App Store.
  - The situation is different on Android, which is open-source.
- I investigate
  - whether such restrictions of Apple negatively affect competition and welfare;
    - I identify factors that determine **whether consumers and developers may be harmed** by such restrictions. Accordingly, whether the DOJ's claim holds is an empirical issue.
  - why Apple does not impose similar restrictions in Asia.
    - A **platform may benefit** from such restrictions only if it has a **larger (initial) consumer-side market share**.
      - This is true for Apple in the U.S. but not in Asia.

Super Apps: How They Differ

- Examples of super apps: WeChat, Alipay, Gojek, Grab.
  - They do not merely serve consumers; they also **host other third-party apps**.
- Consumers can "take" mini apps with them when switching to a new operating system (OS)—the same mini apps are available on super apps there.
  - This **reduces consumers' switching costs**.
- Developers can publish their apps in a single channel and do not need to handle cross-platform data migrations.
  - These **reduce developers' costs of multi-homing**.

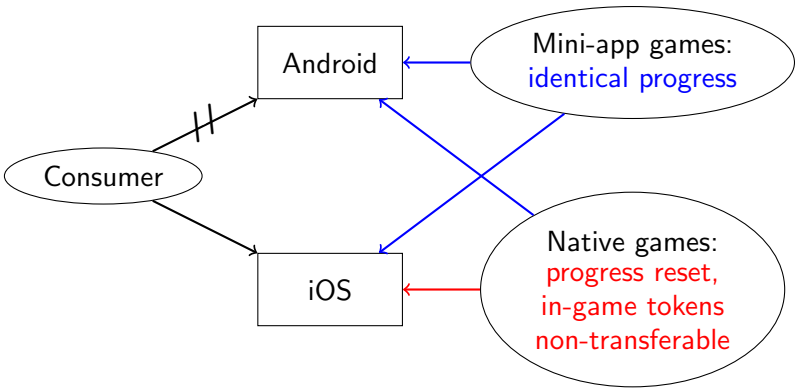
Super Apps: Differences for Consumers—Example: Before Switching

- Consider a consumer who uses a smartphone powered by Android:



Super Apps: Differences for Consumers—Example: After Switching

- When this consumer switches to iPhone:



- Progress reset and non-transferable tokens impose switching costs on consumers.

Modeling Approach

- The smartphone platform market is **two-sided** with **cross-side network effects**.
  - Two sides: consumers and developers.
    — Consumers benefit from the access to more developers, and vice versa.
  - Two competing platforms: operating systems (e.g., iOS and Android).
- Super apps do not act as strategic players.
  - This is comparable to the situation in the U.S. v. Microsoft Corp. (1998) case, where Java and non-Microsoft browsers played a role similar to that of super apps in the U.S. and Plaintiff States v. Apple, Inc. (2024) case.
- Restricting super apps does two things:
  - **Increases consumers' switching costs**.
  - **Decreases developers' multi-homing savings**.

Related Literature

- One-sided market: Viswanathan (2005); Fuentelsaz, Maicas, and Polo (2012); Fabra and García (2015).
  - They yield heterogeneous findings about the price effects of switching costs.
- Two-sided markets with switching costs: Lam (2017); Kim (2017); Giovannetti and Siciliani (2023).
  - The effect of switching costs on consumer surplus can go in either ways.
    - Multi-homing costs or savings are absent from the literature cited above.
- Two-sided markets with multi-homing costs: Jeitschko and Tremblay (2020); Teh et al. (2023).
  - Switching costs are absent.
- Empirical work on two-sided markets with switching and multi-homing costs: Guo et al. (2023).
  - Switching costs and multi-homing costs are
    - on the same side of the market;
    - estimated using multinomial probit model.

Next: Model, Strategies, and Equilibrium

- Motivation and Overview
- Model, Strategies, and Equilibrium
- Comparative Statics, Implications, and Discussions
- Conclusion

Model: Players

- **Platforms**, namely 0 and 1, at end points of Hotelling lines.
- Two groups of users
  - **Consumers** (C):
    - **Horizontal preferences**: $x \sim \mathrm{U}[0,1]$;
    - **Initial platform patronage**: $b \in \{0,1\}$.
      - $f := P(b=0) \in (0,1)$ denotes platform 0's (consumer-side) **initial market share**.
        - Platform 0 is said to be **strong (weak)** if $f > (<) 0.5$ (symmetric for platform 1).
      - Consumers' horizontal preferences are **independent** of their initial platform patronages.
  - **Developers** (D):
    - **Horizontal preferences**: $y \sim \mathrm{U}[0,1]$.

Model: Timing

1. Each platform $i \in \{0, 1\}$ **simultaneously** sets **consumer-side prices** p_i and **developer-side prices** q_i.
   - Each platform's payoff is its **profit**.
   - Each user's payoff is a **baseline payoff** plus **cross-side network effect**, minus the (Hotelling) **transportation cost** and the **price** paid to the platform.
     - Specially, (1) for a **consumer** who switches platform, minus a **switching cost**, and (2) for a **multi-homing developer**, plus a multi-homing **saving**.
2. Observing all prices,
   - each **consumer chooses a platform** to use;
   - each **developer chooses one or two platforms** to operate on.
     - All users choose platforms simultaneously.
     - Consumers can only single-home, while developers can multi-home.
3. Then, the game ends with all parties' payoffs realized.

Key Features that the Model Must Capture

- The smartphone platform market is **two-sided** with **cross-side network effects**.
  - Platforms (OSs) actually sell (the access to) consumers to developers, and vise versa.
  - They **do not separately extract surplus** on each side.
    - Instead, they may subsidize one side to facilitate monetization on the other.
  - Apple's **market power** is also linked to the **relationship between the two sides**.
    - One-sided models cannot capture this.
    - For example, here, Apple uses its **market power over developers**, which **comes from** its unique control over the **access to iOS consumers**, to **affect consumers**' switching costs.
- **Co-existence** of consumers' switching costs and developers' multi-homing savings.
  - The **joint effect** operates through these **two channels, one on each side**.
- **Asymmetric initial market shares** of the platforms.
  - This flexibly captures the sources of platforms' market power in a one-period setting, especially those arising from switching costs.

Modeling Choices

- Pricing on the developer side: The **lump-sum part is strategic**, the **commission part is exogenously fixed**.
  - Translating real-world platform pricing into a formal theoretical model presents substantial complexity:
    - The prices have historically remained fixed.
    - Recent reductions resulted from antitrust litigation rather than from market forces.
  - My choice yields very clean results and intuitions while also allowing for flexibility.
  - In extension, I discuss an alternative that takes both the lump-sum part and commission part as fixed.
- The model is **one-period**.
  - It is equivalent to a two-period setting with sequential entry when letting $f \in \{0, 1\}$.

Model: Consumers' Payoffs

For a consumer with $(x, b) \in [0,1] \times \{0,1\}$ using platform $i \in \{0,1\}$, the payoff is $U_i(x, b)$, with

$$\begin{aligned} U_0(x,b) &= u_0 - sx + c(n_0^D + n_2^D) - p_0 - k\mathbb{1}[b=1], \\ U_1(x,b) &= u_1 - s(1-x) + c(n_1^D + n_2^D) - p_1 - k\mathbb{1}[b=0]. \end{aligned} \tag{2.1}$$

- $u_i > 0$ is consumers' baseline payoff from using platform $i \in \{0,1\}$;
- $s > 0$ is consumers' Hotelling transportation cost parameter;
- $c > 0$ is consumers' cross-side network effect parameter;
- n_i^D is the number of developers that only operate on platform $i \in \{0,1\}$;
- n_2^D is the number of multi-homing developers; and
- $k \geqslant 0$ is each consumer's switching cost.
  - This **switching cost is symmetric**: a consumer incurs the same cost k when switching either to or away from a platform.

Model: Developers' Payoffs

Then, for a single-homing developer with $y \in [0,1]$ operating on platform $i \in \{0,1\}$, the payoff is $V_i(y)$, with

$$\begin{aligned} V_0(y) &= v_0 - ty + dn_0^C - q_0, \\ V_1(y) &= v_1 - t(1-y) + dn_1^C - q_1, \end{aligned} \tag{2.2}$$

and for a multi-homing developer with $y \in [0,1]$, the payoff is

$$V_2(y) = V_0(y) + V_1(y) + r. \tag{2.3}$$

- $v_i > 0$ is developers' baseline payoff from operating on platform $i \in \{0,1\}$;
- $t > 0$ is developers' Hotelling transportation cost parameter;
- $d > 0$ is developers' cross-side network effect parameter;
- n_i^C is the number of consumers that use platform $i \in \{0,1\}$; and
- $r \geqslant 0$ is the amount of each multi-homer's cost saving.
  - Multi-homing always costs more than operating on a single platform. However, it is cheaper than developing for two platforms completely independently.
  - This **cost saving is also symmetric**.

Model: Platforms' Payoffs

For each platform $i \in \{0, 1\}$, its payoff is its profit, which is

$$\pi_i = p_i n_i^C + (q_i + m n_i^C)(n_i^D + n_2^D), \tag{2.4}$$

where $m \geqslant 0$ denotes the **commission** that each platform charges each developer operating on it for each consumer who uses it and is thus accessible to the developers.

— The commission m **is non-negative, exogenous—non-strategic—and identical** across platforms.

Among all of the terms defined above, the prices (p_0, q_0, p_1, q_1) and the market shares $(n_i^C)_{i=0}^1$ and $(n_i^D)_{i=0}^2$ are endogenous and jointly constitute the equilibrium of the game, while all other terms are exogenous.

Strategies

- Platform $i \in \{1, 2\}$: a pair of prices $(p_i, q_i) \in \mathbb{R}^2$ posted to consumers and developers, respectively.
- Consumer: strategy is a function $L_C : [0, 1] \times \{0, 1\} \times \mathbb{R}^4 \to \{0, 1\}$.

  $L_C(x, b, p_0, q_0, p_1, q_1) = i \in \{0, 1\}$ indicates that a consumer with $(x, b) \in [0, 1] \times \{0, 1\}$ chooses to use platform i given prices (p_0, q_0, p_1, q_1).
- Developer: a function $L_D : [0, 1] \times \mathbb{R}^4 \to \{0, 1, 2\}$.

  $L_D(y, p_0, q_0, p_1, q_1) = i \in \{0, 1, 2\}$ indicates that given prices (p_0, q_0, p_1, q_1), a developer with $y \in [0, 1]$ chooses to

  1. operate on platform i only (when $i \in \{0, 1\}$) or
  2. multi-home (when $i = 2$).

Equilibrium

- Equilibrium concept: SPE.
- In the equilibrium,
  1. given platforms' equilibrium prices and other users' equilibrium strategies, each user optimally chooses platform(s) to use or operate on to maximize his or her payoff, and
  2. given users' equilibrium strategies and the other platform's equilibrium prices, each platform optimally sets its prices to maximize its profit.
- I solve for the equilibrium by working backward from the second stage of the game and focus on interior solutions under full market coverage.
- With this focus, and under certain assumptions on parameter values, the model possesses a unique equilibrium $[(p_0^*, q_0^*), (p_1^*, q_1^*), L_C^*, L_D^*]$ with market shares $(n_i^{C*})_{i=0}^1$ and $(n_i^{D*})_{i=0}^2$.

Next: Comparative Statics, Implications, and Discussions

- Motivation and Overview
- Model, Strategies, and Equilibrium
- Comparative Statics, Implications, and Discussions
  - Main focus: Effects of restrictions on super apps on profits and welfare.
  - Because such restrictions affect both consumers' switching costs and multi-homers' cost savings, I
    - first separately investigate the effects of the above two, and
    - then put them together to consider the (joint) effect of the restrictions.
- Conclusion

Comparative Statics: Profits—k

Proposition 1 (Effects of Consumers' Switching Costs on Profits)

An increase in consumers' switching costs k increases (decreases) the equilibrium profit of a platform if and only if the platform is strong (weak) on the consumer side.

- When k increases, a **strong platform** ends up **hosting more users** on both sides of the market despite its **higher consumer-side price**.
- Though it may has incentive to decrease its developer-side price to facilitate exploiting consumers, this never reverses the overall effect of k on its equilibrium profit.
- The weak platform experiences exactly the opposite outcome of the strong one.

Comparative Statics: Profits—r

Proposition 2 (Effects of Multi-homers' Cost Savings on Profits)

An increase in multi-homers' cost savings r increases the equilibrium profit of each platform.

- When r increases, both platforms **host more developers** despite their **higher developer-side prices**—some developers are driven from single-homing to multi-homing. Their equilibrium consumer-side market shares are unaffected.
- Though they both reduce their consumer-side prices, hoping to facilitate exploiting developers, this never reverses the overall effect of r on the equilibrium profits.

Comparative Statics: Profits—Restrictions on Super Apps I

Recall that the effects of restrictions on super apps are via two channels:
(an increase in) k and (a decrease in) r.

Corollary 1 (Effects of Restrictions on Super Apps on Profits)

In response to restrictions on super apps,

1. *the equilibrium profit of a platform that is not strong on the consumer side decreases; and*
2. *the direction of change in the equilibrium profit of a platform that is strong on the consumer side is ambiguous: the profit increases (decreases) when the restrictions increase consumers' switching costs to a significantly greater (lesser) extent than they decrease multi-homers' cost savings.*

- When a platform is not strong, the restrictions harm it via both channels.
- For a strong platform, the overall effect is positive when the restrictions increase k to a significantly greater extent than they decrease r.

Comparative Statics: Profits—Restrictions on Super Apps II

- Figure 1 shows how the direction of the effect of restrictions on super apps on the strong platform's equilibrium profit depend on
  - consumers' horizontal differentiation; and
    - x-axis: s.
  - the extent to which the restrictions increase consumers' switching costs by Δk and decrease multi-homers' cost savings by Δr.
    - y-axis: $\Delta r/\Delta k$.

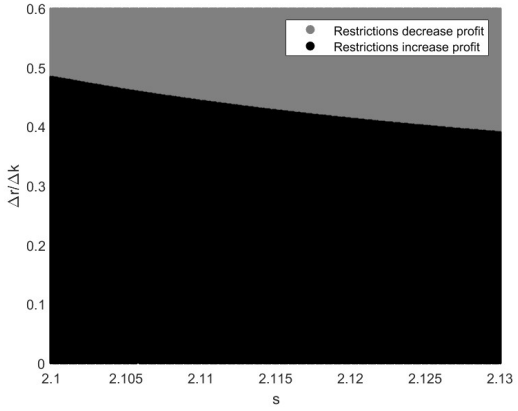


Figure 1: Direction of the Effect of Super App Restrictions on Strong Platform Profit, Degree of Consumers' Horizontal Differentiation $s \in [2.1, 2.13]$

Comparative Statics: Welfare—k I

Proposition 3 (Effects of Consumers' Switching Costs on Consumer Surplus)

When there is an increase in consumers' switching costs k, the equilibrium consumer surplus

1. *decreases when the degree of consumers' horizontal differentiation is sufficiently large relative to consumers' switching costs and the magnitudes of users' network effects;*
2. *may increase when the degree of consumers' horizontal differentiation is not sufficiently large.*

1. When **consumers are very horizontally differentiated**, most **consumers' payoffs decrease in** k.
   - — Exception: Those who initially patronize the weak platform and end up using it. However, these consumers are so few that the overall effect of k on the equilibrium consumer surplus is negative.
   - — One reason: Consumers are so strongly tied to the platforms that are closer to them that many of them opt to switch despite rising costs and do not sufficiently aggregate on a single platform.

Comparative Statics: Welfare—k II

2. When **consumers are not very differentiated**, if one of the two platforms is sufficiently strong, the payoff of each consumer that ends up using the strong platform increases in k.
   - Intuition: A sufficiently large number of consumers and developers may cluster on the stronger platform, generating **substantial cross-side network effects**.

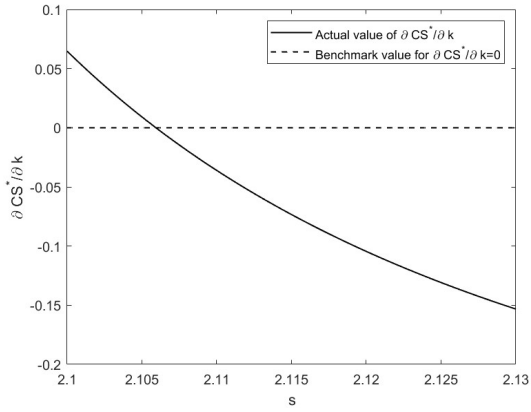


Figure 2: Effect of Consumers' Switching Costs k on Consumer Surplus, Degree of Consumers' Horizontal Differentiation $s \in [2.1, 2.13]$

Comparative Statics: Welfare—k III

Proposition 4 (Effects of Consumers' Switching Costs on Developer Surplus)

When there is an increase in consumers' switching costs k,

1. *the equilibrium developer surplus increases if and only if the strong platform serves more developers in equilibrium than the weak platform; and*
2. *the resulting change in the equilibrium developer surplus diminishes in magnitude and ultimately vanishes as consumers become sufficiently horizontally differentiated.*

- k affects developers via developer-side prices and consumer-side market shares.
- **Developers benefit** from an increase in k if and only if it **single-homes on the strong platform**.
  - This is via the channel of cross-side network effects.
- When consumers are sufficiently horizontally differentiated, the above two channels become too insensitive for changes in k to generate any substantial effect on developers.

Comparative Statics: Welfare—r

Proposition 5 (Effects of Multi-homers' Cost Savings on Welfare)

An increase in multi-homers' cost savings r

1. *increases the equilibrium consumer surplus; and*
2. *decreases the equilibrium developer surplus.*

- Consumers benefit from consumer-side price reductions.
- Developers are harmed by increases in developer-side prices.

Comparative Statics: Welfare—Restrictions on Super Apps I

Corollary 2 (Effects of Restrictions on Super Apps on Welfare)

In response to restrictions on super apps,

1. *when consumers are sufficiently horizontally differentiated, the equilibrium consumer surplus decreases, while the equilibrium developer surplus increases;*
2. *when consumers are not very horizontally differentiated and the restrictions increase consumers' switching costs to a significantly greater extent than they decrease multi-homers' cost savings, the equilibrium consumer surplus may increase, and the equilibrium developer surplus may decrease.*

- When consumers are sufficiently horizontally differentiated, they are harmed by the restrictions via both channels, and the effect on developers via r outweighs that via k.
- Otherwise,
  - consumers prefer both a higher level of k and a higher level of r,
  - the effect on developers via k is non-negligible
    - — the direction of this effect depends on the equilibrium developer-side market shares.

Comparative Statics: Welfare—Restrictions on Super Apps II

- Figure 3 shows how the direction of the effects of restrictions on super apps on the strong platform's equilibrium profit and equilibrium consumer surplus depend on s and $\Delta r/\Delta k$.
  - When the strong platform benefit from the restrictions, it is unclear whether consumers are harmed or not.

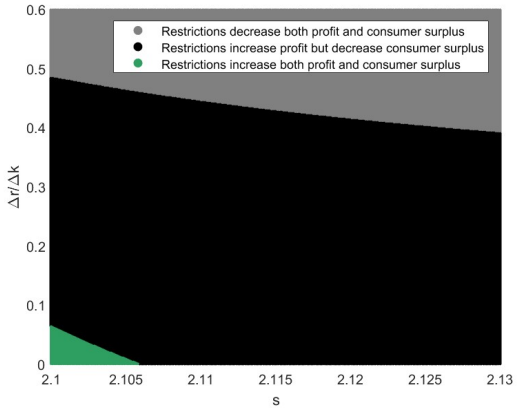


Figure 3: Direction of the Effect of Super App Restrictions on Strong Platform Profit and Consumer Surplus, Degree of Consumers' Horizontal Differentiation $s \in [2.1, 2.13]$

Summary: Antitrust-Related Implications Supporting the DOJ's Claim

- A **strong platform** (OS) may have incentive to **impose restrictions** on super apps, while a **weak platform never** does.
  - Potential antitrust concern: Apple's U.S. restrictions on super apps may **harm rivals**.

- When **consumers perceive the platforms as highly differentiated**, they are **harmed** by such restrictions.
  - Another potential antitrust concern: such restrictions of Apple may harm consumers.

Summary: Antitrust-Related Implications Opposing the DOJ's Claim

- **Consumers may not be harmed** by such restrictions when
  1. they view the **platforms as sufficiently close substitutes** and
  2. these restrictions
     - **significantly increase their adaptation costs** when switching to a new platform but
     - **marginally heighten developer workloads** associated with platform-specific maintenance.

- This weakens the DOJ's consumer-harm claim
  - especially because the second condition mirrors the environment in which Apple benefits from restricting super apps in the U.S.

Discussion: Extension—Fixed Developer-Side Prices

- In reality, in the smartphone platform market, **both lump-sum fees and commissions charged to developers do not change much**.
  - This questions the effect of multi-homing savings r on developer surplus:

    When r increases, developers are shown to be harmed by increases in developer-side prices. However, what if developer-side prices are actually unaffected by r?
- Thus, I extend the model by taking **developer-side prices** q_0 and q_1 **fixed**, so that each platform's strategy is simply its consumer-side price.
- Unlike in the baseline model, I find **developer surplus to increase in** r in this extension.
  - The other key findings from the baseline model remain robust to this extension.

Discussion: Implications from the Extension

- **Developers may not be harmed** by such restrictions when
  1. **consumers** view the platforms as **sufficiently close substitutes**,
  2. these restrictions
     - **significantly increase consumer adaptation costs** when switching to a new platform but
     - **marginally heighten developer workloads** associated with platform-specific maintenance, and
  3. **the strong platform hosts more apps** (developers) than the weak platform.

  - This weakens the DOJ's developer-harm claim if and only if all of the above conditions hold.
    - Especially, the second condition mirrors the environment in which Apple benefits from restricting super apps in the U.S.

Concluding Remarks

- I model the smartphone platform market as a two-sided market with cross-side network effects to analyze the joint effect of consumers' switching costs and developers' multi-homing cost savings under restrictions on super apps.
- Restrictions on super apps **may benefit users** when implemented.
  - Factors that determine **whether such restrictions may benefit** consumers and developers—and thus warrant closer empirical examination:
    - Whether consumers view platforms as highly differentiated or close substitutes.
    - The degree to which restrictions on super apps (1) increase consumer adaptation costs upon switching platforms and (2) heighten developer workloads associated with platform-specific maintenance.
    - The number of apps operate on each platform.
    - Whether developer-side pricing exhibits flexibility or rigidity.
  - The DOJ has **argued on conceptual grounds that Apple behaves anticompetitively by restricting super apps**. Nevertheless, I show that **reaching a definitive conclusion requires further quantitative analysis** that accounts for the factors identified above.
- One key **reason why Apple restricts super apps** in the U.S.: **iOS is strong** in the U.S. smartphone platform market.
- Such restrictions always **harm Apple's weak rivals** in the U.S.

Possible Future Extensions

- Introducing same-side network effects.
  - In one-sided markets, they tend to reduce the harm of switching costs.
- Introducing utility values of super apps.
  - There may be no effect via this channel if restriction on super apps symmetrically affect the direct utilities on both platforms.
  - If a platform's restrictions only reduce the direct utilities on itself, platforms would face reduced incentives to implement them.

Key Distinctions from Other Apple Antitrust Cases

- In prior antitrust litigation involving Apple, such as Epic Games, Inc. v. Apple Inc. (2021), the focus centered on Apple's exercise of market power against developers rather than on the consumer side of the platform.
  - In this context, only the developer side is directly relevant, making a single-sided model sufficient.
- In Epic Games, Inc. v. Apple Inc. (2021), Apple imposed restrictions primarily to prevent developers from bypassing its 30% commission.
  - This sharply contrasts with its suppression of super apps, which Apple restricts even though developers are willing to comply with standard revenue-sharing requirements.